Sound Waves

Exam code: 0625 & 0972

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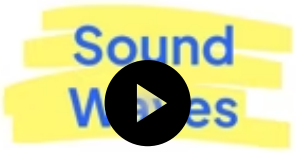

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Describing sound

- Sound waves are produced by **vibrating** sources
- Sound waves are Longitudinal
  - So a **medium** is needed to transmit sound waves
  - This means if there are no molecules, such as in a **vacuum**, then the sound can't travel through it
- When a sound wave comes into contact with a solid, the longitudinal wave vibrations are **transferred** to the solid
  - For example, sound waves can cause a drinking glass to vibrate
  - If the glass vibrates too much the movement causes the glass to shatter

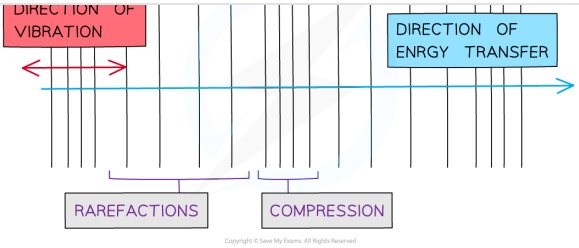
Representing longitudinal waves

- Longitudinal waves are usually drawn as several lines to show that the wave is moving **parallel** to the direction of energy transfer
  - Drawing the lines closer together represents the **compressions**
  - Drawing the lines further apart represents the **rarefactions**

Representing a longitudinal wave with compression and rarefaction

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



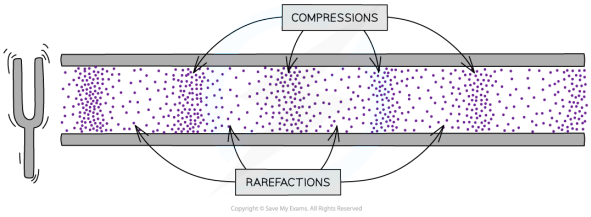
Longitudinal waves are represented as sets of lines with rarefactions and compressions

Compression & rarefaction

Extended tier only

- Longitudinal waves consist of compressions and **rarefactions**:
  - A **compression** is a region of higher density i.e. a place where the molecules are bunched together
  - A **rarefaction** is a region of lower density i.e. a place where the molecules are spread out

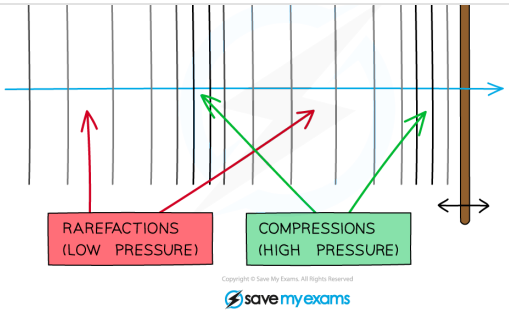
Compression and rarefaction



Sound is a longitudinal wave consisting of compressions and rarefactions – these are areas where the pressure of the air in the pipe varies with the wave

- These compressions and rarefactions cause changes in pressure, which vary in time with the wave
  - Therefore, sound is a type of **pressure wave**
- When the waves hit a solid, the variations in pressure cause the surface of the solid to **vibrate** in sync with the sound wave

Compression and rarefaction in contact with a solid



When sound waves hit a solid, the fluctuating pressure causes the solid to vibrate

Examiner Tips and Tricks

When describing compressions and rarefactions, make sure to use the correct terms. It is best to refer to them as regions of high and low densities of particles instead of the particles are more 'bunched up' or 'far apart', as this is too vague and not very scientific!

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